Physical Therapy in Sport xxx (2010) 1–13

Contents lists available at ScienceDirect

Physical Therapy in Sport

journal homepage: www.elsevier.com/ptsp

Masterclass

A return-to-sport algorithm for acute hamstring injuries

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ARTICLE INFO

Article history:
Received 11 February 2010
Received in revised form
9 July 2010
Accepted 12 July 2010

Keywords:
Muscle strain
Hip extension
Optimum angle
Eccentric intervention
H/Q ratio

ABSTRACT

Acute hamstring injuries are the most prevalent muscle injuries reported in sport. Despite a thorough and concentrated effort to prevent and rehabilitate hamstring injuries, injury occurrence and re-injury rates have not improved over the past 28 years. This failure is most likely due to the following: 1) an over-reliance on treating the symptoms of injury, such as subjective measures of "pain", with drugs and interventions; 2) the risk factors investigated for hamstring injuries have not been related to the actual movements that cause hamstring injuries i.e. not functional; and, 3) a multi-factorial approach to assessment and treatment has not been utilized. The purpose of this clinical commentary is to introduce a model for progression through a return-to-sport rehabilitation following an acute hamstring injury. This model is developed from objective and quantifiable tests (i.e. clinical and functional tests) that are structured into a step-by-step algorithm. In addition, each step in the algorithm includes a treatment protocol. These protocols are meant to help the athlete to improve through each phase safely so that they can achieve the desired goals and progress through the algorithm and back to their chosen sport. We hope that this algorithm can serve as a foundation for future evidence based research and aid in the development of new objective and quantifiable testing methods.

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1. Introduction

Hamstring muscle strains are the most prevalent muscle injuries reported in sport. Epidemiology studies have revealed that hamstring injuries alone account for between 6 and 29% of all injuries reported in Australian Rules football, rugby union, soccer, basketball, cricket and track sprinters (Brooks, Fuller, Kemp, & Reddin, 2005a; 2005b; Croisier, 2004; Garrett, 1996; Meeuwisse, Sellmer, & Hagel, 2003; Orchard & Seward, 2002; Woods et al., 2004). In addition to the prevalence of hamstring injuries, frustration can be intensified by prolonged symptoms, poor healing responses and a high risk of re-injury at a rate of 12–31% (Croisier, 2004; Woods et al., 2004). Even more troubling is the fact that hamstring injury and re-injury rates have not improved over the last 28 years (Ekstrand & Gillquist, 1983; Hägglund et al., 2009). The constant re-injury rates are especially troubling as re-injuries are significantly more severe than initial injuries (Croisier, 2004; Werner et al., 2009; Woods et al., 2004). In addition, previous injury has constantly been found to be one of the greatest risk factors for future injury. These findings suggest that traditional hamstring prevention and rehabilitation programs have not been effective.

Traditionally, the criteria for an athlete to return-to-sport after an acute hamstring injury include a general post-injury timeline, isolated isokinetic strength testing, and subjective feedback from the patient and coaching/medical staff (Clanton & Coupe, 1998; Drezner, 2003; Heiderscheit, Sherry, Silder, Chumanov, & Thelen, 2010; Hoskins & Pollard, 2005a, 2005b; Hunter & Speed, 2007; Petersen & Holmich, 2005; Worrell, 1994). There are seven published studies on the treatment and management of acute hamstring injuries (Clanton & Coupe 1998; Drezner, 2003; Heiderscheit et al., 2010; Hoskins & Pollard, 2005a, 2005b; Hunter & Speed, 2007; Petersen & Holmich, 2005; Worrell, 1994). Each of these studies has identified three basic phases of rehabilitation: 1) the acute phase; 2) the sub-acute/rehabilitation phase; and, 3) the functional phase (see Tables 1 and 2). As can be seen in Table 1, the criteria for progressing to the second and third phases are determined by subjective measures and/or a post-injury timeline. However, clinicians should be aware of the potential gap between patients perceived and actual sport readiness. For example, in anterior cruciate ligament (ACL) injury studies, patient's subjective scores did not significantly correlate with quantifiable strength and functional measures (Neeb, Aufdemkampe, Wagener, & Mastenbroek, 1997; Ross, Irrgang, Denegar, McCloy, & Unangst, 2002). Only three of the seven studies mention an objective measure (i.e. isokinetic strength asymmetries) for progressing from the third phase back-to-sport

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doi:10.1016/j.ptsp.2010.07.003

Table 1
Previous literature on the criteria for progression through a return-to-sport rehabilitation.

Study	Acute phase criteria	Sub-acute phase criteria	Functional phase criteria
Worrell (1994)	Inflammation down	None	Pain free sports movements
Petersen and Holmich (2005)	Inflammation down	None	Pain free sports movements
Hoskins and Pollard (2005a, 2005b)	None	None	Pain free sports movements <10% Isokinetic strength w/un-injured
Clanton and Coupe (1998)	<1 week	Pain free full ROM	Pain free sports movements
Hunter and Speed (2007)	Roughly 5 days post-injury	Full ROM Generate force Control eccentric movements	Pain free sports movements
Drezner (2003)	None	None	Pain free sports movements <10% Isokinetic strength w/un-injured
Heiderscheit et al. (2010)	Normal walking stride without pain	Full strength (5/5) without pain during prone knee flexion (90°) manual strength test	< 5% Isokinetic Functional Ratio w/un-injured
	Very low speed jog without pain		4 consecutive repetitions of maximum effort manual strength test (90° and 15°)
	Pain free isometric contraction against sub-maximal (50–70%) resistance during prone knee flexion (90°) manual strength test	Pain free forward and backward jog, moderate intensity	Full ROM without pain Pain free sports movements

Key: ROM = range of motion.

(Drezner, 2003; Heiderscheit et al., 2010; Hoskins & Pollard, 2005a, 2005b). However, it has been shown that concentric strength levels do not always decrease during isokinetic concentric testing and hamstring-to-quadriceps (H/Q) ratios are not affected after hamstring injuries (Bennell et al., 1998; Brockett, Morgan, & Proske, 2004; Worrell, Perrin, Gansneder, & Gieck, 1991). Heiderscheit et al. (2010) is the most current and thorough of the hamstring management studies. Several detailed exercises are presented through a three phase progression (i.e. acute, regeneration and functional phase). However, this article also fails to provide any insight beyond subjective, ROM or isokinetic criteria for progressing an athlete back-to-sport.

We propose that a multi-factorial approach to rehabilitating hamstring injuries is needed, which includes reliable, objective and quantifiable criteria (clinical and functional) in order to determine how and when to progress a patient through each phase of a return-to-sport rehabilitation program. This algorithm is based on the various risk factors for hamstring injuries, and incorporates the

Table 2
Previous literature on the management of acute hamstring injuries.

Study	Acute phase treatment	Sub-acute phase treatment	Functional phase treatment
Worrell (1994)	RICE	Isolated strengthening (isometric then concentric then eccentric)	Sport specific movements
	NSAIDS Resume normal gait pattern Active knee flexion and extension Stretching	Eccentric "Swing Catches" Static and advanced stretching Swimming/pool exercise, cross-training	Jog to run to sprint progression
Petersen and Holmich (2005)	RICE NSAIDS (short term only)	Isometrics at various angles Isolated stretching Isolated strengthening (i.e. isometric then concentric then eccentric)	Sport specific movements Jog to run to sprint progression
Clanton and Coupe (1998)	RICE NSAIDS	Stretching Isolated strengthening (isometric then concentric then eccentric)	Sport specific movements Jog to run to sprint progression
	Pain free stretching Normal gait	Swimming/pool exercise, cross-training	Normal strengthening and stretching
Hunter and Speed (2007)	Movement in pain free ROM RICE	Stretching Isolated strengthening (isometric then concentric then eccentric)	Sport specific movements Jog to run to sprint progression
	NSAIDS	Nordic hamstring, ficks, wobbles	
Drezner (2003)	Immobilization RICE	Stretching Isolated strengthening (isometric then concentric then eccentric)	Sport specific movements Jog to run to sprint progression
	NSAIDS	Biking, Swimming, Cross-Training	Normal strengthening and stretching
Hoskins and Pollard (2005a, 2005b)	Immobilization < 1 week Cryotherapy/RICE	Stretching Isolated strengthening (isometric then concentric then eccentric) SIJ Manipulation	Sport specific movements Jog to run to sprint progression
Heiderscheit et al. (2010)	RICE Avoid pasive and active lengnts Core (side, prone, front planks)	Lunge walk with trunk rotation Rotation body bridge Grapevine jog	Skip Rotation body bridge with dumbbells Lunge walk with trunk rotation with dumbbells
	Stationary bike	Single limb balance windmill touches without weight	Sport specific movements
	Side step Single limb balance	Supine bent knee bridge with walk outs Stationary bike	Stationary bike

Key: NSAIDS = Non-steroidal anti-inflammatory drugs; RICE = Rest, Ice, Compression, Elevation.

Please cite this article in press as: Mendiguchia, J., Brughelli, M., A return-to-sport algorithm for acute hamstring injuries, Physical Therapy in Sport (2010), doi:10.1016/j.ptsp.2010.07.003

current literature on biology of muscle injury and repair. The severity or injury shouldn't affect the different phases of the algorithm, but would make it more difficult to achieve the criteria to advance through each phase. It should be noted that this algorithm has not yet been validated. However, each objective criterion in the model has shown to be reliable in the literature and clinical rationale is provided. We hope that this clinical commentary can inspire critical evaluation of the model (see Fig. 1), and lead to the development of further reliable, objective and quantitative measures encompassing a multi-factorial approach to rehabilitating acute hamstring injuries.

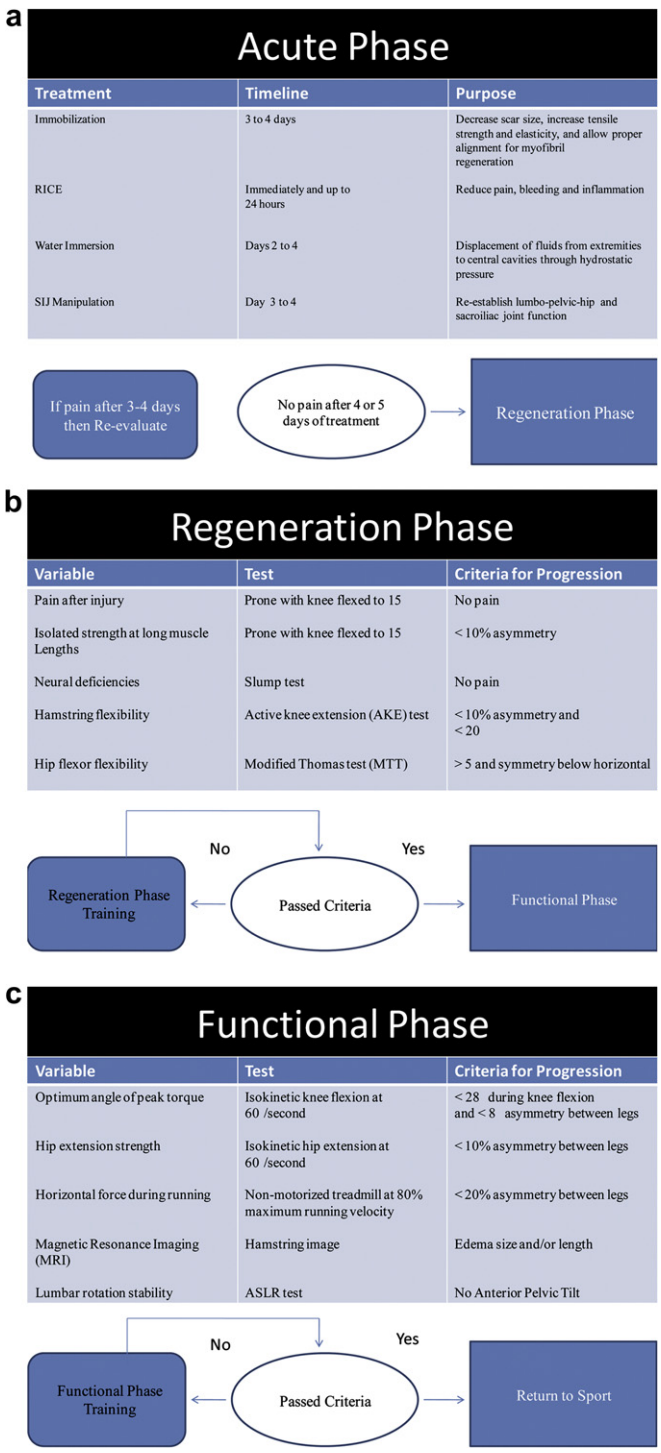

Fig. 1. a. The acute phase of the return-to-sport algorithm. b. The sub-acute/regeneration phase of the return-to-sport algorithm. c. The functional phase of the return-to-sport algorithm.

2. Hamstring algorithm phases

A rehabilitation program should take an athlete through a combination of low-risk and high-demand movements. The aim of training should be to develop functional abilities of the athlete while minimizing the risk of injury. Objective criteria should be used to progress an athlete through each phase of rehabilitation i.e. the acute phase, the sub-acute/regeneration phase, and the functional phase (see Fig. 1). The ultimate goal of the hamstring return-to-sport algorithm is to identify and treat deficits (i.e. neuromuscular and biomechanical deficits) that influence performance and re-injury. This algorithm incorporates objective and functional criteria (statics and dynamics) for progressing through each phase of rehabilitation, and incorporates the most recent training methods for developing/re-developing normal neuromuscular and biomechanical function.

3. Acute phase

The goals for the acute phase include: 1) preventing re-ruptures to the injured site; 2) preventing excessive inflammation and scar tissue; 3) increase tensile strength, adhesion and elasticity of the new granulation tissue; 4) reduce interstitial (i.e. between cells) fluid build-up; and, 5) detect and treat any lumbo-pelvic dysfunction (see Fig. 1a)

3.1. Mobilization vs. immobilization

Experimental research has shown that if slight mobilization is carried out immediately after injury, larger scar tissue evolves and the myofibril branches that penetrate the scar tissue are impaired (Jarvinen, Jarvinen, Kaariainen, Kalimo, & Järvinen, 2005; Jarvinen et al., 2007). Also, further tissue damage is common at the site of injury if mobilization is begun too soon (Jarvinen, 1975, 1977). Conversely, early immobilization can prevent excessive scar tissue and re-ruptures (Jarvinen, 1975, 1977; Jarvinen & Lehto, 1993; Jarvinen, Einola, & Virtanen, 1992). Early immobilization allows for new development of granulation tissue with appropriate tensile strength and elasticity (Jarvinen et al., 2005). However if immobilization is carried out for too long, detrimental effects have been reported which can affect proper healing. (Jarvinen et al., 2005). Excessive immobilization has been shown to induce excessive fibrosis, atrophy of the muscle fibers, and loss of strength and elasticity (Jarvinen, 1975). Based on experimental findings, Jarvinen and co-authors (Jarvinen et al., 2007) recommended early immobilization after an acute hamstring injury (i.e. 3–4 days), followed by active mobilization in the regeneration/sub-acute phase. Experimental data has shown that beginning active mobilization after early immobilization enhances the penetration of myofibril branches through the granulation tissue, decreases the size of the permanent scar, increases tensile strength and elasticity, and allows for proper alignment and regeneration of myofibrils (Jarvinen et al., 2005; Jarvinen et al., 2007).

3.2. Cryotherapy and hydrotherapy

The RICE principle (*i.e. rest, ice, compression and elevation*) has been shown to be very practical and is often used to reduce pain and bleeding. In experimental research, ice has been shown to reduce inflammation and the size of the hematoma after injury, and thus reduce permanent scar tissue (Jarvinen et al., 2007; Swenson, Sward, & Karlsson, 1996). Compression has been shown to reduce

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intramuscular blood flow to the injured site. However it is debatable whether compression should be applied in the first 24 h. It has been recommended that ice and compression should be alternated as this combination has been shown to reduce intramuscular temperature (3–7°) and blood flow (50%) (Thorsson, Lilja, Dahlgren, Hemdal, & Westlin, 1985). However, no evidence of an optimal mode or duration of RICE exists, (Bleakley, McDonough, & MacAuley, 2004) and it has been suggested that more hamstring specific trials are needed (Hoskins & Pollard, 2005a).

Water immersion has gained popularity for its effects on increasing intracellular intravascular fluid shifts, reduction of muscle oedema, and increased cardiac output without energy expenditure which is thought to increase blood flow and transportation of nutrient and waste production throughout the body (Wilcock, Cronin, & Hing, 2006). Unfortunately, the effects of water immersion are only being studied on the physiology of recovery after exercise and no studies are investigating the effects of muscle injury and repair. As the body is submerged in water, a compressive force is applied to the body called hydrostatic pressure. This pressure causes the fluids in the body to become displaced from the extremities to the central cavity of the body (Lollgen, von Nieding, Koppenhagen, Kersting, & Just, 1981). The amount of pressure that acts on the body is depended on the depth of submersion, not on the total amount of water. At hip level submersion, the fluids are displaced from the lower extremities (i.e. higher pressure area) to the thoracic region (i.e. lower pressure area) (Lollgen et al., 1981; Wilcock, et al., 2006). The potential benefits of water immersion on muscle strain injuries include: preventing inflammation and oedema, transporting blood from interstitial and intramuscular space to intravascular space, reducing the permanent scar tissue, and aid in the transportation of waste products away from the injured site (Wilcock et al., 2006). In addition to RICE 24-h post-injury, Wilcock and co-authors (Wilcock et al., 2006) recommended cold water immersion for 10 min at 25 degrees, which is thought to increase movement of interstitial-intravascular fluids. Water immersion should be performed without passive or active movement for the following 2–3 days. We recommend no more than 2–3 water immersions up to hip level per day. It should be noted that heat and contrast therapy should be avoided during this phase due to a possible increase in inflammation.

3.3. Sacroiliac joint manipulation

The sacroiliac joint (SIJ) links the two lower extremities with the spine, which effectively transfers loads from spine to the legs. It has been proposed that any SIJ dysfunction could lead to leg asymmetries during functional movements, altered gait patterns, early hamstring activation and loss of pelvic stability (Cibulka, Sinacore, Cromer, & Delitto, 1998; Herzog & Conway, 1994; Hungerford, Gilleard, & Hodges, 2003; Mason, Dickens, & Vail, 2007). Specifically the contribution of biceps femoris, via its insertion through sacrotuberous ligament and attachment to thoracolumbar fascia, has been shown to increase sacroiliac joint stiffness (Van Wingerden, Vleeming, Buyruk, & Raissadat, 2004). Therefore any pelvis position change or neuromuscular dysfunction can alter the load transfer from the spine to the legs increasing the risk of injury. Moreover, altered pelvic function due to a past history of groin or osteitis pubis has been suggested to be a significant risk factor for hamstring injury (Verrall, Slavotinek, Barnes, Fon, & Spriggins, 2001).

Manipulation of sacroiliac joint has been purposed and used successfully in the literature as a tool to re-establish the lumbopelvic function (Cibulka, Rose, Delitto, & Sinacore, 1986; Hoskins & Pollard, 2005a). One randomized study showed improved hamstring strength after SIJ manipulation compared to a control group with no SIJ manipulation (Cibulka et al., 1986). These findings suggest that any alterations of the sacroiliac joint function can affect hamstrings mechanical behaviour (Cibulka et al., 1986). Hoskins and Pollard (2005b) attributed a successful correction of anterior pelvic tilt, after SIJ manipulation, with the successful rehabilitation of two Australian Rules football players with previous hamstring injuries. However, it should be noted that more research is needed in this area to validate the effectiveness of SIJ manipulation.

3.4. Non-steroidal anti-inflammatory drugs (NSAIDS)

Non-steroidal anti-inflammatory drugs (NSAIDS) are commonly recommended for acute muscle strain injuries, especially in the short term as their long-term use seems to be detrimental to the regenerating skeletal muscle. NSAIDS work through the inhibition of prostaglandin production. It is prostaglandin that serves as one of the mediators in the inflammatory process, but reductions in prostaglandin levels do not always correlate with beneficial results in muscle injury models (Mishra, Friden, Schmitz, & Lieber, 1995). In fact, it has been shown that NSAIDS have detrimental effects on muscle repair as they reduced local prostaglandin E2 (Dinoprostona) concentration, which is one of the biggest source for satellite cell synthesis (Mikkelsen, Helmark, Kjaer, & Langberg, 2008). Satellite cells are transformed into new muscle cells during the repair phase after injury.

There are currently no random controlled studies that have reported beneficial or superior effects of NSAIDS compared to analgesics or placebo on acute muscle strain injuries. For example, Reynolds, Noakes, Schwellnus, Windt, and Bowerbank (1995) studied the effect of NSAIDs compared to placebo in combination with physiotherapy for the treatment of acute hamstring injuries and found no additional benefit with NSAIDs over standard physiotherapy alone. Similarly, Warren, Gabbe, Schneider-Kolsky, and Bennell (2008) did not show any significant effect of NSAIDs use or not in recovery time or as re-injury predictor in AFL players that suffered hamstring strains, but underline the importance of reducing the pain to move through the rehabilitation process. For this reason Rahusen, Weinhold, and Almekinders (2004) has suggested that the routine use of NSAIDS for muscle injuries may need to be critically evaluated because low-cost and low-risk analgesics may be just as effective. Despite the universal acceptance for NSAIDS usage for acute hamstring injuries (see Table 2), further research is needed on the safety and effectiveness of NSAIDS before they can be recommended for practical use.

If the symptoms caused by the injured muscle persist more than 5 days after the trauma, it may be necessary to reconsider the existence of more extensive tissue damage or intramuscular hematoma that might require special attention. If there are no problems after 5 days, the athlete can progress to the sub-acute phase (see Fig. 1a).

4. Sub-acute/regeneration phase

The goals of the sub-acute/regeneration phase include: 1) improve overall core stability; 2) improve strength and symmetry, and reduce pain during prone isometric isolated (hamstring) contractions at 15° of knee flexion; 3) improve hamstring flexibility of both legs; 4) improve hip flexor flexibility of both legs; and, 5) improve neuromuscular control.

4.1. Core stability

Despite the popularity and interest in the core in the last decade, "core stability" is one of the most misused terms in the literature. It has been incorrectly used synonymously and interchangeably with balance, core strength, hip strength and spine stability. In this

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paper, the "core" musculature will be referred to as that musculature that surrounds and inserts in the lumbo-pelvic region (i.e. a total of 29 muscles) (Bliss & Teeple, 2005). These muscles act synergistically to stabilize the trunk and hip, and significantly contribute to the stability of the knee joint. Core stability depends on the relationship between the passive structures, the ligaments, vertebral facets and the active neuromuscular controllers. Optimal recruitment, strength and endurance of the 29 muscles (attached to the pelvis) are necessary to maintain and restore joint (core) homeostatic stability in response to internal or external forces from expected or unexpected perturbations. This occurs through all planes of motion and despite changes in the center of gravity.

Many articles suggest that deficits in hip strength are related to a high risk of injury in the lower extremities. For example, Leetun, Ireland, Willson, Ballantyne, and Davis (2004) reported that a lack of core stability, defined as a decreased hip strength in female athletes, can predict injuries in the lower extremities. Although the hip is part of the core, care must be taken to not use hip strength as synonymous with core stability. Even today it is unknown how hip strength is used in stabilizing manner. It remains unclear how strength affects core stability and vice versa. Quantitative analysis suggests that 10% of maximum voluntary contraction (MVC) of abdominal co-contraction may be sufficient to achieve spine stability during normal movements (McGill, 2002). More recently it has been reported that stability is achieved in the first 25% of MVC (Brown, Vera-Garcia, & McGill, 2006). Therefore, it is possible that feedback control or muscle endurance may be more important than strength for reducing the risk of injury. In other words, an athlete can be very strong in their core musculature, but have poor core stability due to poor motor patterns, reflex pathways or muscular endurance. More research is needed in this area before definitive conclusions can be made.

Many authors have speculated that low back pain could be a risk factor for acute hamstring injuries. Various studies have reported reduced trunk muscle force, (Taimela & Harkapaa, 1996) endurance, (Biedermann, Shanks, Forrest, & Inglis, 1991) different activation patterns, (Hodges, Cresswell, Daggfeldt, & Thorstensson, 2000; Reeves, Cholewicki, & Silfies, 2006) disturbed postural control, (Luoto, Taimela, Hurri, & Alaranta, 1999) altered trunk propioception, (Taimela & Harkapaa, 1996) and hip strength (Nadler, Malanga, DePrince, Stitik, & Feinberg, 2000) and reduced gluteal activation (Kankaanpaa, Taimela, Laaksonen, Hänninen, & Airaksinen, 1998; Leinonen, Kankaanpaa, Airaksinen, & Hänninen, 2000) after low back pain (McGill, 2007). Thus low back pain should be considered a source of instability and treated in athletes with previous hamstring injuries.

Recently, core stability has been linked with hamstring injury (Chumanov, Heiderscheit, & Thelen, 2007; Mason, Dickens, & Vail 2007; Sherry & Best, 2004; Thelen, Chumanov, Sherry, & Heiderscheit, 2006) and has become a cornerstone of different rehabilitation and treatment programs (Hewett, Myer, & Ford, 2006; Mascal, Landel, & Powers, 2003; Myer, Ford, McLean, & Hewett, 2006; Sherry & Best, 2004). Sherry and Best (2004) found that a group of athletes who performed a core stability rehabilitation program suffered significantly less hamstring injuries in comparison with a group of athletes that performed only isolated strength and stretching. For the remainder of this section, all of the risk factors for hamstring injuries that have been linked with core stability have been included: hamstring strength at long lengths, hamstring flexibility, neural tension, hip flexor flexibility, and gluteus maximus strength and activation.

4.2. Strength at long muscle lengths

Pain and deficits in hamstring strength are common after acute injuries, especially at long muscle lengths. There are two main reasons why this variable should be assessed before allowing an injured athlete to the functional phase. First, hamstring injuries are thought to occur when the muscle is activated beyond their optimum length (length at which the greatest toque is able to be generated by the muscle) and it has been proposed that weakness at longer muscle lengths (i.e. during hip flexion and/or knee extension) is a risk factor for future injury. Second, the biceps femoris has been shown to be activated at longer lengths (i.e. 15–30 degrees of knee flexion), compared to the semitendinosus and semimembranosus muscles (i.e. 90–105 degrees of knee flexion) (Onishi, Yagi, Oyama, Akasaka, Ihashi, & Handa, 2002). In addition, the long head of the biceps femoris is the most commonly injured hamstring muscle (72–80% of all hamstring injuries) (Askling, Tengvar, Saartok, & Thorstensson, 2007; Connell et al., 2004; Hoskins & Pollard, 2005a; Hunter & Speed, 2007; Koulouris, Connell, Brukner, & Schneider-Kolsky, 2007; Woods et al., 2004). Thus it is important to know how the muscle is functioning at longer than optimum muscle lengths.

Hamstring strength at long lengths can be assessed during isometric contractions at 15 degrees of knee flexion in a lying prone position (see Fig. 2) (Warren, Gabbe, Schneider-Kolsky, & Bennell, 2008). Hand held dynamometers have been shown to have very good to excellent inter-rater, intra-rater, and inter-session reliability during lower extremity testing with appropriate stabilization and tester strength (Kornberg & Lew, 1989; Krause, Schlagel, Stember, Zoetewey, & Hollman, 2007; Lu, Hsu, Chang, & Chen, 2007). Most recently, Kelln, McKeon, Gontkof, & Hertel (2008) reported intra-rater and inter-session reliability of ICC = 0.83–0.95 during lying prone knee flexion with the knee flexed at 90 degrees. In order to maximize stabilization and leverage, Warren et al. (2008), recommended the following position for measuring hamstring strength with a hand held dynamometer: the subject lays on the ground (not on a table) and the tester bends over the patient's ankles with arms extended and shoulders over his/her hands (see Fig. 2). This position will prevent the patient from being able to overpower the tester, which has shown to reduce reliability of hand held dynamometry (Lu et al., 2007). In order to pass this criterion, the athlete needs to achieve a leg asymmetry of less than 10% in hamstring strength as proposed by Warren et al. (Warren et al., 2008).

All of the studies outlined in Tables 1 and 2 recommend performing strengthening of the hamstring with a progression from; isometric contractions at various angles to concentric to eccentric contractions. These authors make the argument that if eccentric muscle contractions are started first, then greater forces will be created which could cause further damage. However, during the sub-acute rehabilitation phase of acute muscle injuries serious consideration should be given to performing repetitive concentric

Fig. 2. Hamstring strength assessment at 15 degrees of knee extension in the prone position.

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contractions at short muscle length including, but not limited to: cycling and rowing. It has been shown that cyclist (i.e. perform repeated concentric muscle contractions at short muscle lengths), produce peak tension at shorter muscle lengths in comparison to runners (i.e. perform eccentric and concentric contractions) in the rectus femoris (Herzog, Guimaraes, Anton, & Carter-Erdman, 1991; Savelberg & Meijer, 2003). It is likely that the hamstrings may also adapt to concentric based training at short muscle lengths. Three recent studies have shown that isometric and concentric training at short lengths can shift the optimum length of tension development to shorter muscle lengths (Blazevich, Horne, Cannavan, Coleman, & Aagaard, 2008; Kilgallon, Donnelly, & Shafat, 2007; Mjolsnes, Arnason, Osthagen, Raastad, & Bahr, 2004). A shorter than normal optimum length has been suggested as a risk factor for future hamstring injury (Arnason, Andersen, Holme, Engebretsen, & Bahr, 2008; Brockett et al., 2004; Brooks, Fuller, Kemp, & Reddin, 2006). Furthermore, in a rehabilitation study by Sherry and Best (2004), a group of athletes performed cycling training for 10–15 min per workout at low to moderate intensities in addition to hamstring stretching and concentric strengthening, and reported a 70% re-injury rate. Conversely, eccentric training at long muscle lengths has been shown to increase the optimum length of tension development, and significantly reduce hamstring injury rates (Arnason et al., 2008; Askling, Karlsson, & Thorstensson, 2003; Brooks et al., 2005a; Gabbe, Branson, & Bennell, 2006a; Proske, Morgan, Brockett, & Percival, 2004). Kubo et al. (2006) and Philippou, Bogdanis, Nevill, & Maridaki (2004) reported that the optimum lengths increased after isometric training at long muscle lengths, but not at short isometric lengths. Most recently, Brughelli, Cronin, and Nosaka (2009) investigated optimum lengths (knee flexion and knee extension) between trained cyclists and Australian Rules football players. The ARF players provided an appropriate model for comparison as they perform a mixture of training methods and muscle contractions, where the cyclists only perform concentric muscle contractions at short muscle lengths. It was reported that the cyclists had a shorter optimum length during knee flexion (6.1°) and knee extension (4.3°), although peak torque and muscle thickness were not significantly different ($p < 0.05$). Alternative methods to cycling should be considered for improving and maintaining aerobic endurance after an acute hamstring injury.

4.3. Hamstring flexibility

Two recent prospective studies on hamstring injuries have reported that injured elite soccer players had significantly reduced hamstring flexibility in comparison with un-injured elite soccer players (Bradley & Portas, 2007; Witvrouw, Danneels, Asselman, D'Have, & Cambier, 2003). Two older retrospective studies reported that previously injured athletes had significantly lower hamstring flexibility in comparison to un-injured athletes (Jönhagen, Nemeth, & Eriksson, 1994; Worrell et al., 1991). Furthermore, Worrell et al. (1991) reported an asymmetry between legs (i.e. injured and non-injured leg) in hamstring flexibility during rehabilitation after injury, with the injured leg being significantly less flexible.

Hamstring flexibility should be restored soon after injury. However, the treatment for hamstring flexibility should avoid stress the sciatic nerve (i.e. increasing neural tension). Neural tension has been identified as a risk factor for future hamstring injury (Kujala, Orava, & Jarvinen, 1997; Turl & George, 1998). Traditional static stretching techniques that involve the combination of cervical flexion, hip flexion and dorsiflexion have been shown to increase neural tension (Butler & Wolkenstein, 1991; Kornberg & Lew, 1989; Turl & George, 1998). Such stretches include variations of the hurdlers stretch (see Fig. 3) and toe touches. In order to assess hamstring flexibility and avoid neural tension Hunter and Speed (2007) recommend the active knee extension test (AKE) as opposed to the straight leg raise. The inter-rater and inter-session reliability of the AKE test have ranged between ICC = 0.92–0.96 (Gabbe, Finch, Wajswelner, & Bennell, 2004). The AKE is a measure of hamstring flexibility taken at 90 degrees of hip flexion. At the point of maximal active knee extension (or onset of pain), the angle between the vertical and the tibia can be recorded by an inclinometer. For improving hamstring flexibility and avoiding neural tension, Hunter and Speed (2007) proposed dynamic physiological mobilization stretches. To determine if internal or external rotation is needed, a very simple test has been proposed i.e. "taking off the shoe" test (TOST) (Zeren & Oztekin, 2006). The TOST has been shown to have a sensitivity, specificity and accuracy of 100% when compared with ultrasound images diagnosing biceps femoris muscle strains (Zeren & Oztekin, 2006). In addition, *soft tissue mobilization techniques* proposed by Hooper et al. (2005) can be used in this phase. However, a more dynamic and functional approach may be desired to increasing hamstring flexibility, such as the "ball go and back" (see Fig. 4 and b) that involves hip frontal stability and neuromuscular control, which has been shown associated with hamstring strain (Cameron, Adams, & Maher, 2003).

Fig. 3. The hurdlers stretch.

4.4. Neural tension

As mentioned previously, neural tension has been proposed as a risk factor for hamstring injury. Several studies have confirmed that 14–19% of all hamstring injuries reported are without any MRI-confirmed structural muscle damage and linked with the neuromeningeal structures (Verrall et al., 2001). A recent study found that as high as 45% of hamstring injuries were without damage, suggesting no local muscle pathology (Gibbs, Cross, Cameron, & Houang, 2004). In other scenarios, hamstring injuries and neural tension may be associated. Turl et al. found that 57% of Rugby players suffering from hamstring injuries presented neural tension at same AKE-measured flexibility values between injured and control subjects (Turl & George, 1998). Neural tension has been defined as abnormal physiological and mechanical response in nervous system structures when the normal range of movement and capabilities are exceeded (Gallant, 1998). Both tensile and compressive forces can affect neural tissue and produce damage to the neural system. Normal neural tissue is not painful both at rest and during motion. Neural tension has been described as sharp burning pain, which symptoms are not generally associated with

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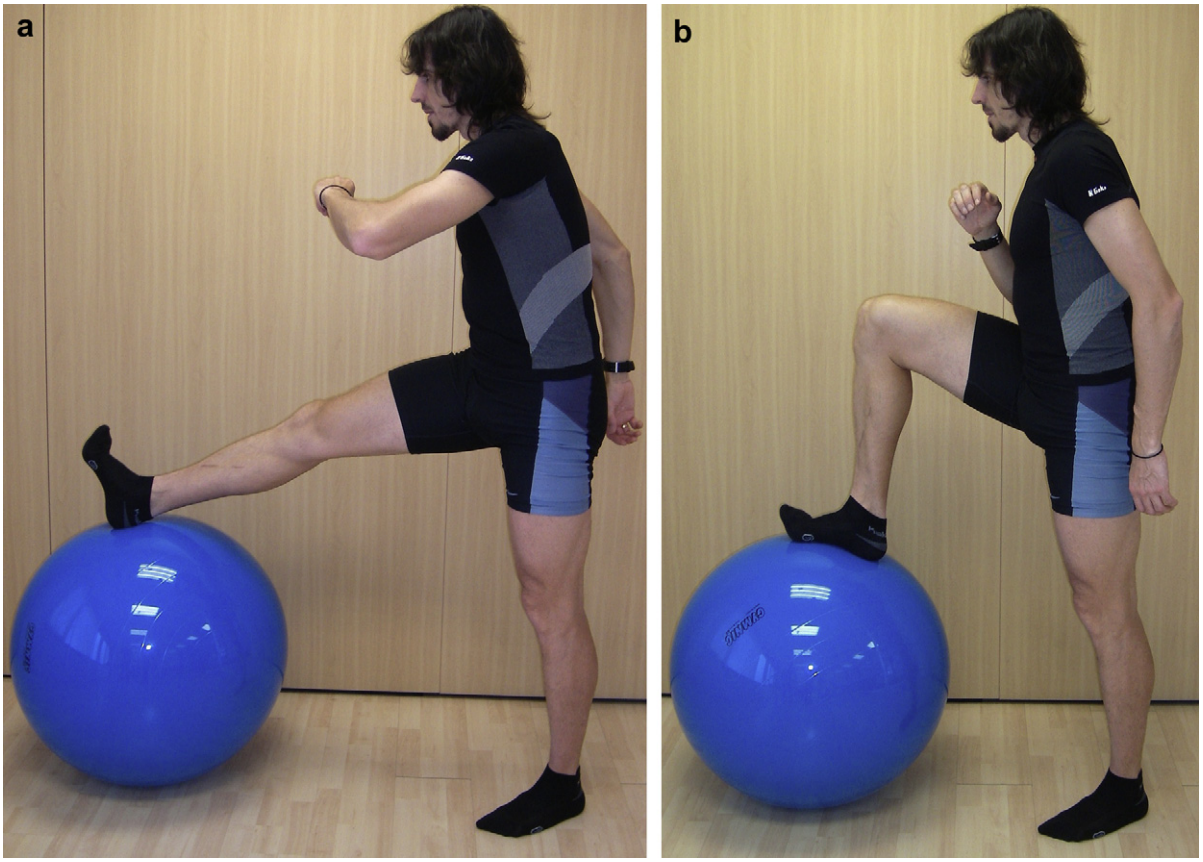

Fig. 4. a–b. The "ball and go" exercise.

muscle strain injuries. An athlete might also report a dull pain located deep in the buttocks or posterior thigh associated with prolonged sitting (Butler, 1991).

Differentiation of hamstring muscle tightness and neural tension can be achieved with the "Slump Test" (Butler, 1991). The active slump test assesses pain-sensitive neuromeningeal structures that have been suggested as a potential source of pain in the posterior thigh in hamstring injuries. In the sitting position the athlete is instructed to clasp his hands behind his back, to tuck his chin onto his chest and to slump bringing his shoulders towards his hips with full cervical, thoracic and lumbar flexion. Next, full active dorsiflexion of the foot of the injured leg was requested and the athlete actively extends his knee until he/she feels a stretch or hamstring pain. The athlete will then be asked to extend his neck to a neutral position and describe the change in sensation that occurred in the hamstring. The test is considered positive if the athlete's original hamstring pain was decreased, and then reproduced with cervical flexion (Butler, 1991; Gallant, 1998). The athlete might complain of a burning or stinging sensation at the end range of the motion or report sharp pain likely to be located in the popliteal fossa, adjacent to the fibular head or in the lumbar spine (Gallant, 1998) as opposed to a stretching sensation. For a positive slump test Butler (Butler, 1991) proposed a treatment protocol based on specific release and tension techniques.

4.5. Hip flexor flexibility

Another risk factor that has been identified in the literature for future hamstring injuries is hip flexor flexibility. Gabbe, Bennell, and Finch (2006b) reported that older Australian Rules football (ARF) players were at greater risk for hamstring injuries and had reduced hip flexor flexibility (measured with the modified Thomas test) in comparison to younger ARF players. Since an athlete's age is considered one of the greatest risk factors for hamstring injuries, it was concluded that hip flexor flexibility was also a risk factor. Furthermore, Chumanov et al. (2007) studied the effects of running velocity and the influence of individual muscles on hamstring stretch. The activation of the illiopsoas (stance leg), greatly increased the stretch of the hamstrings of the swing leg. At maximum running velocity, the hip flexors induced a 20 mm increase in contralateral Biceps Femoris stretch. This increase of 20 mm stretch is comparable to the decrease in stretch induced by the hamstrings themselves (Chumanov et al., 2007). Recently, Franz, Paylo, Dicharry, Riley, and Kerrigan (2009) reported that subjects with decreased hip extension mobility consistently compensated with increased anterior pelvic tilt during the stance phase of both walking and running. It was shown that patients with hip flexion contractures (i.e. limited hip extension) an increase in stride length during running is commonly achieved by compensating by increasing anterior pelvic tilt and lumbar extension during late stance. Furthermore, it has been speculated that an increase in anterior pelvic rotation, due to tight hip flexors, could increase the length of the activated hamstring muscles and thus increase the risk of acute injury (Chumanov et al., 2007; Gabbe et al., 2006b; Schache, Bennell, Blanch, & Wrigley, 1999; Schache, Blanch, Rath, Wrigley, Starr, & Bennell, 2001).

For assessing hip flexor flexibility we recommend the Modified Thomas Test (MTT) (Harvey, 1998). For the modified Thomas test, the subject will sit on the end of the table and lay back into a supine position. The athlete will then pull both knees to their chest. The

athlete will hold the contralateral hip in maximal flexion with the arms, while the tested limb will be lowered toward the floor. The axis of the goniometer will be placed over the greater trochanter, with the fixed axis directed vertically. The moveable arm of the goniometer will be pointed toward the lateral knee joint line, representing the line of the femur. The tester will then assess the hip angle relative to the horizontal. A negative angle represents flexion above the horizontal and a positive angle represents extension below the horizontal. Gabbe et al. (2006b) reported that for each 1° increase in the MTT, the risk of hamstring strain is increased by 15% in athletes older than 25 years old. To increase hip flexor flexibility, proprioceptive neuromuscular facilitation (PNF) stretching and a progression to dynamic functional stretches has been purposed by Stuart McGill (McGill, 2002).

4.6. Gluteus maximus strength and activation

The main functions of the gluteus maximus (GM) during running are to control trunk flexion of the stance leg, decelerate the swing leg and extend the hip (Lieberman, Raichlen, Pontzer, Bramble, & Cutright-Smith, 2006; Muckle, 1982; Novacheck, 1998). The timing and magnitude of electromyography (EMG) patterns of the GM and hamstring have been shown to be similar (Jönhagen, Ericson, Nemeth, & Eriksson, 1996; Simonsen, Thomsen, & Klausen, 1985). Therefore, any alteration in GM activation, strength, or endurance places greater demand on the hamstrings to control hip extension of the stance leg and decelerate the leg during the swing phase. The gluteus maximus provides powerful hip extension when sprinting, and the hamstrings help to transfer the power between the hip and knee joints. For improving GM activation, strength, and endurance the following recommendations have been proposed: teaching good motor patterns and isolating the GM from hamstrings, bridges with both legs and progression to one leg, and finally reintegrate the GM with the hamstrings with exercises such as single-leg deadlifts and lunges (Brughelli & Cronin, 2007b; Farrokhi, Pollard, Souza, Chen, Reischl, & Powers, 2008).

5. Functional phase

The goals of the functional phase include: 1) increasing the optimum length of the hamstrings; 2) decrease leg asymmetries in optimum length; 3) decrease leg asymmetries in concentric hip extension; 4) decrease leg asymmetries in horizontal force production during running; and, 5) improve torsional capabilities.

5.1. Optimum angle of peak torque

Skeletal muscles have an optimum length for producing peak tension. Muscle strain injuries are thought to occur when activated muscles are lengthened to greater than optimal lengths (Brockett et al., 2004; Brooks et al., 2006; Proske et al., 2004). The hamstring muscles are actively lengthened during hip flexion and knee extension, which occur simultaneously during the late swing phase in running (i.e. as the air borne leg swings forwards). A recent retrospective study has identified the optimum length as a risk factor for injury. Brockett, Morgan, and Proske (1999) measured the optimum lengths in athletes with previously injured hamstrings. One leg served as the experimental leg (i.e. previously injured hamstring) and the other leg served as the control leg (i.e. un-injured hamstring). The previously injured hamstring produced peak tension at 12.7 degrees less than the un-injured hamstring (i.e. shorter optimum length).

Isokinetic dynamometers have shown to be mechanically valid and reliable in regards to torque, velocity and position (Drouin, Valovich-mcLeod, Shultz, Gansneder, & Perrin, 2004). Brockett et al. (1999) reported that the optimum angle of peak torque can be reliably calculated at an angular velocity of 60 degrees per second.

It has been argued that hamstring injuries can be reduced if this optimum length can be increased through training (Brockett, Morgan, & Proske 2001; Brockett et al., 2004). The only form of training that has been shown to consistently increase the optimum length of tension development has been eccentric exercise (for a recent review see Brughelli & Cronin, 2007a). Furthermore, the only form of training that has consistently been shown to reduce hamstring injury rates is eccentric training (Arnason et al., 2008; Askling et al., 2003; Gabbe et al., 2006a). For increasing the optimum length eccentric exercises that actively lengthen the hamstrings with either hip flexion, knee extension or a combination of both have been proposed by Brughelli and Cronin (2007b). Since hamstring injuries occur proximally and distally from the insertion (Askling, Tengvar, Saartok, & Thorstensson, 2008) both locations should be trained eccentrically. Brughelli and Cronin (2007b) suggest using a more functional approach to exercise design in comparison with the previous literature, that involves closed-chain and multi-joint exercises.

It should be noted that the optimum length is always measured during concentric contractions at relatively slow angular velocities. Despite these limitations, optimum length has been shown to be decreased after injury (Brockett et al., 2001; Brughelli et al., 2010), and eccentric exercise has been shown to both increase optimum length (Brockett et al., 2001; Brughelli et al., 2010, Clark, Bryan, Culpan, & Hartley, 2005) and decrease injury rates (Arnason et al., 2008; Askling et al., 2003; Gabbe et al., 2006a). Furthermore, optimum length has been shown to be consistent amongst contraction type. Thus, optimum length is an important variable for assessing injury risk and monitoring progression of an eccentric based intervention.

5.2. Strength imbalances

One of the proposed risk factors for acute hamstring injuries is muscle weakness during concentric and/or eccentric contractions (Croisier, 2004; Croisier, Ganteaume, Binet, Genty, & Ferret, 2008). Muscle weakness has been assessed with one of two methods: 1) comparing the peak torque values of the knee extensors (during concentric contraction) with their antagonistic muscle group i.e. the knee flexors (during concentric or eccentric contraction); and, 2) comparing the peak torque values of the one leg with the contralateral leg during knee flexion. Both methods have produced conflicting findings in prospective and retrospective studies (Bennell et al., 1998; Brockett et al., 2004; Croisier et al., 2008; Heiser, Weber, Sullivan, Clare, & Jacobs, 1984; Lieholm, 1978; Sugiura, Saito, Sakuraba, Sakuma, & Suzuki, 2008; Orchard, Marsden, Lord, & Garlick, 1997; Worrell et al., 1991; Yeung, Suen, & Yeung, 2009).

However, there is consistent evidence to suggest that eccentric peak torque during knee flexion is reduced after an acute hamstring injury. Sugiura et al. (2008) recently reported that eccentric peak torque was significantly decreased in six sprinters who sustained an acute hamstring injury over a 12 month period. Croisier et al. (Croisier et al., 2008; Croisier, 2004) have reported that mixed eccentric and concentric H/Q ratio disorders could be used to identify subjects who were at risk for future injury in a prospective study, and detecting 70% of subjects who suffered hamstring injuries in a retrospective study. However, these studies did not report if the injuries occurred in the same leg or in the contralateral leg. Dauty, Potiron-Josse, and Rochcongar (2003) reported that mixed concentric/eccentric H/Q ratio disorders could also identify athletes who have had previous injury, but the ratio could not predict new hamstring injuries. Very interestingly, over a period of 12 months the injured subjects suffered new injuries in the leg with

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better mixed eccentric/concentric H/Q ratios. In other words, the opposite leg (i.e. previously un-injured leg) was injured. Eccentric peak torque should be regarded as a risk factor for future hamstring injury, however both legs should be considered "at risk". In addition, caution should be used any time maximal eccentric contractions are being performed at long muscle lengths, especially with athletes recovering from an acute injury. Orchard et al. (2001) reported a hamstring muscle strain injury, confirmed by MRI, during eccentric isokinetic testing at 180 degrees per second. In addition to the Nordic hamstring exercise, Brughelli and Cronin (2007b) proposed several functional eccentric hamstring exercises for increasing eccentric strength.

5.3. Hip extension strength

Recently, Suguira et al. (2008) reported that elite sprinters who sustained acute hamstring injuries had reduced concentric hip extension strength. Since the majority of hamstring injuries occur more proximally towards the hip joint during sprinting and since these injuries take longer to recover, (Askling, Saartok, & Thorstensson, 2006; Askling et al., 2008) it is important to assess concentric strength of the gluteus as they help to extend the hip. The assessment of concentric hip extensor strength involves an isokinetic dynamometer with the subject positioned in a stranding position (Sugiura et al., 2008). The subject will perform concentric contractions at 60 degrees per second (Sugiura et al., 2008).

In order to increase concentric hip extensor strength a variety of concentric step-ups and lunges have been proposed in the literature by Jönhagen, Ackermann, and Saartok (2009) and Farrokhi et al. (2008). Both exercises should be initiated from a static position and the contribution of the back leg should be minimized. The emphasis of the exercise should be placed on the front leg to lift the body. The exercises can be overloaded with extra resistance or increased velocity (i.e. jumping from a static position). These exercises are intended to increase hip extensor strength and overall gluteal strength.

5.4. Leg asymmetries in horizontal force

In addition to assessing the mechanical capabilities of the lower body during open chain isokinetic testing, it is important to also assess the functional capabilities during a closed-chain and multi-joint movement. Yu, Queen, Abbey, Liu, Moorman, and Garrett (2008) recently reported that the hamstrings undergo an eccentric contraction during the late stance phase as well as during the late swing phase of over-ground running. Since it has been proposed that hamstring injuries may occur during the swing phase and stance phase in running, it is important to assess any deficits in force production during the stance phase of running.

In a recent study by Brughelli, Cronin, Mendiguchia, Kinsella, and Nosaka (2009) Australian Rules football players with previous hamstring injuries were compared (i.e. kinetic and kinematic variables) with non-injured athletes during running on a non-motorized force treadmill. It was reported that the previously injured athletes had significant leg asymmetries in horizontal force production during running (45.9%) (see Fig. 5), but not in vertical force production. We feel that there are two possible explanations for the asymmetries in horizontal force production but not vertical force production after hamstring injury: 1) increased anterior pelvic tilt and reduced leg extension during the stance phase: and, 2) the proximal to distal transfer of power between joints is altered during the stance phase. It has been suggested that relationships might exist between certain kinematic parameters of the lumbo-pelvic-hip complex and running related injuries (Franz et al., 2009; Schache, Blanch, & Murphy, 2000; Schache, Blanch, Rath, Wrigley, & Bennell, 2005). Specifically, hamstring injuries have been associated with increased pelvic rotation and reduced hip extension during running (Franz et al., 2009; Gabbe et al., 2006b; Schache et al., 2005). Excessive anterior pelvic rotation during running is thought to be caused by a decrease in hip flexor flexibility (Franz et al., 2009; Gabbe et al., 2006b; Schache et al., 1999; Schache et al., 2001). The reliance on anterior pelvic rotation during the stance phase, and possibly lumbar flexion, is thought to decrease hip extension range of motion (Franz et al., 2009). The decreased hip extension would most likely decrease horizontal force production during running. Although horizontal force would be decreased during this period, vertical force and contact time would not be expected to be altered by pelvic rotation and/or hip extension.

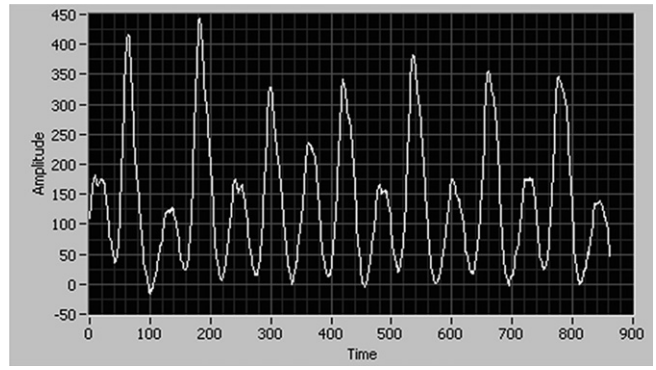

Fig. 5. Horizontal forces in injured ARF players during running demonstrating significant contralateral leg deficits of 45%.

The second possible explanation for the leg asymmetries during running after a hamstring injury could be due to an alteration in the proximal to distal transfer of power between joints. Upon landing and throughout the first half of the stance phase, the hamstrings help to extend the hip and keep the knee joint flexed. An early increase in leg extension would lead to an increase in vertical velocity of the CM, which would interfere with the horizontal acceleration of the CM (Jacobs, Bobbert, & van Ingen Schenau, 1993, 1996). The hamstrings help to delay the explosive leg extension and allow the body to rotate over the ankle. During the second half of the stance phase, net power (joint moment and angular velocity) is thought to be transferred from the hip joint to the knee joint. Thus the bi-articulate muscles contribute to the transfer of net power from proximal to distal joints, which allows for an efficient conversion of body segment rotations (during first half of stance phase) into the translation of the CM in the horizontal direction (Jacobs et al., 1993, 1996). If the hamstrings are injured, it could be speculated that this sequence would be disrupted and horizontal force production would be decreased. Conversely, vertical force production would not be expected to be affected by a disruption in the proximal to distal sequence of net power transfer as vertical force is mainly dependent upon mass, gravity, vertical velocity and leg spring stiffness during human running (Blickhan, 1989; McMahon & Cheng, 1990). More research is needed on the effects of hamstring injuries on running kinetics and the proximal to distal sequence of net power transfer between joints.

Horizontal force production can be measured directly from the force plate or from a load cell tethered to the athlete. Previous research has reported that non-motorized force treadmills are reliable and valid for both kinetic and kinematic variables in comparison to over-ground running (Chelly & Denis, 2001; Hughes, Doherty, Tong, Reilly, & Cable, 2006; McKenna & Riches, 2007; Sirotic & Coutts, 2008; Tong, Bell, Ball, & Winter, 2001). For increasing horizontal force production of the injured leg and decreasing this asymmetry, we recommend unilateral and bilateral

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exercises that allow the subjects to produce strength and power in the horizontal direction.

5.5. Lumbar rotation capabilities

Recently, core stability and hamstring injury has been linked in the literature. For this reason, we recommend that torsional capabilities of the trunk should be assessed in this phase of the algorithm where the athlete will be exposed to kicking, sprinting and changes of directions activities. The majority of clinicians do not have the instrumentation that is necessary for calculating spine stability. The ASLR Test has recently been used as a screen of lumbar spine stability to assess the control of lumbar rotational movements in the transverse plane (Liebenson, Karpowicz, Brown, Howarth, & McGill, 2009). This test can be used to follow up athlete's improvements in torsional capabilities. Good control without anterior pelvic tilt is required before the athlete can progress through the algorithm. Anterior pelvic tilt typically accentuates the lumbar lordosis and can be a sign of poor stabilization of the pelvis by the abdominal muscles.

Sherry and Best (2004) reported that a group of athletes who performed progressive trunk and agility exercises suffered fewer hamstring injuries than a group that performed isolated hamstring strengthening and stretching exercises. The trunk stabilization exercises appeared to be effective and capable of changing the altered motor patterns derived from low back pain or core instability. Treatment for core stability and specifically torsional capabilities of the trunk includes exercises that progress from static with planks in the regeneration phase to more dynamic exercises in the functional phase (Heiderscheit et al., 2010; McGill, 2007; Sherry & Best, 2004).

5.6. Imaging techniques

Image techniques, such as ultrasound (US) and magnetic resonance imaging (MRI), have been used to diagnose and monitor hamstring injuries. The general advantages of US consist of low-cost, availability and their non-invasive nature. However, there are clear disadvantages of US being highly operator-dependent and unable to image bone. In contrast, MRI has traditionally served as an objective standard for confirming the presence of injury and presents superior tissue contrast resolution. However, MRI equipment is relatively expensive and difficult to use as a daily tool in the field. Due to the availability and nature of US, it can be useful in following healing processes, and it provides essential feedback to both the athlete and clinician. MRI may have a more significant role in the management of muscle injury in elite athletes, specifically where acute decisions regarding imminent participation in sport are necessary.

Recent studies have shown that the location and extent of abnormalities (e.g., oedema and haemorrhage) on MRI not only confirm the presence and severity of initial muscle fiber damage but can also provide a reasonable estimate of the rehabilitation period (Askling et al., 2007; Slavotinek, Verrall, & Fon, 2002). Some controversial data has been published with respect to re-injury rates. Koulouris et al. (2007) found a strong correlation association between MRI images (i.e. length of a strain) with hamstring injury recurrence rates, which suggests that MRI may be able to identify athletes at risk of re-injury. However other studies have not found significant correlations between MRI and hamstring injury rates (Gibbs et al., 2004). Connell et al. (2004) reported that no prognostic significance was attributed to either the location of the injury (proximal or distal) regardless of which muscle was involved, or the type of tear (musculotendinous or myofascial) (Connell et al., 2004).

The role of MRI and US in the assessment of appropriate timing to safely return-to-sports training has also been investigated in the literature, which include: measurement of the separation between the normal margins (percentage of muscle involvement), the filling of the haemorrhagic cavity by a fibrotic tissue, and the assessment of the magnitude of the scar formation (proportional to the risk of recurrent injury) have been used to determine the healing status (Peetrons, 2002; Van Holsbeeck & Introcaso, 2001). However, it is very difficult to assess a safe return to play exclusively based in US and MRI parameters. There are two main reasons for this difficulty:

1. The image findings related to muscle strains may persist after resolution of clinical symptoms. More research and evidence is needed to resolve this question.
2. No image technique is able to reflect structural and mechanical properties of the injured muscle.

Therefore, the information from image techniques should be used in conjunction with other objective tests. This approach may increase the success a return-to-sport rehabilitation program and reduce the risk of re-injury. Future research should investigate the relationships between specific biomechanical tests and image techniques.

6. Conclusions

Return-to-sport rehabilitation programs that only rely on subjective measures such as "pain free movements", may result in deficits in neuromuscular control, strength, flexibility, ground reaction force attenuation and production, and lead to asymmetries between legs during normal athletic movements. These deficits and deficiencies could persist into sport practice and competition, and ultimately increase the risk of re-injury and limit athletic performance. A criteria based approach to rehabilitation, that includes objective and quantitative tests has the potential to identify deficits and address them in a systematic progression (i.e. algorithm) during the stages of returning to sport. Ultimately, the algorithm approach may lead to a successful return-to-sport with a reduced risk of injury. However, it should be noted that further research is needed (i.e. prospective, retrospective and training studies) in order to validate the criteria based progressions in each phase.

Conflict of interest statements

None.

Ethical approval

None.

Acknowledgments

We thank Eduard Alentorn - Geli MD for the stimulating discussion related to this study.

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Please cite this article in press as: Mendiguchia, J., Brughelli, M., A return-to-sport algorithm for acute hamstring injuries, Physical Therapy in Sport (2010), doi:10.1016/j.ptsp.2010.07.003

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Please cite this article in press as: Mendiguchia, J., Brughelli, M., A return-to-sport algorithm for acute hamstring injuries, Physical Therapy in Sport (2010), doi:10.1016/j.ptsp.2010.07.003